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Characterisation of nano-engineered dental tools

An analysis will be done on different dental tools and burs, these tools are used in the dental industry to shape and cut tooth materials. In this project an understanding of the tools and their surface properties will be investigated to learn how to increase tool life and cutting speeds. In the dental industry there are many different types of tools which aid dental specialists with inspecting and treating teeth and gums. A brief description of the most common tools used include the mouth mirror which is a frequently used tool usually used in conjunction with other dental tools to provide a clear image of areas of the mouth otherwise obstructed. A tool often combined with the mouth mirror are Dental probes. Dental probes are split into two different types; periodontal probes and sickle probes and are generally employed to test how healthy the teeth and the surrounding area are and whether further action needs to be taken. Finally, dental handpieces are crucial devices used within the dental industry with several different types including high speed or low speed, the handpieces are combined with dental burs to cut away teeth and hard tissue. Other dental tools include:

- Excavators
- Burnishers
- Pluggers
- Curettes
- Dental lasers
- Forceps
- Chisels

→ this phrase is too long.

Different types of Burs and their surface properties are going to be the focus of this project each one will be characterised using scanning electron microscopes and other instruments and then compared with burs that have a titanium or carbon nano-engineered films on the surface, using instruments including the scanning electron microscope and x-ray diffraction machine. ✓

→ which ones?

Dental burs are used to cut away tooth material or restorative material such as porcelain, they have two parts to them the small drill piece and the shank which is the longer part of the bur (Sampson-Field, n.d.), the shank connects the bur up to the handpiece, it comes in an assortment of lengths depending on the handpiece it is being attached to. Burs are generally composed of diamond, tungsten carbide or steel with carbide being the most common, which in turn impacts the price of the burs depending on the different material and size of the bur with diamond being the most expensive then tungsten carbide and steel being the most economical. Burs are not currently thrown away after a single use but are instead sterilized and then reused however 'most bur and diamond companies are now offering low cost disposable alternatives. (A.Molinari, n.d.)'. Dental burs have a surface topography that makes cleaning and sterilization hard to achieve and therefore can take longer than other dental tools to clean. Theories to improve dental burs have been discussed with coating tungsten carbide burs in diamond being a popular opinion.

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→ Time is missing.

Regular meetings with my supervisor have been organised for every Thursday to discuss how the project is going and the progress that I am making. To complete this project, I am going to send out letters to different dental labs and ask for any used or spare dental burs that I can

chapters and sections are missing

→ very good initiative.

use. Figure.1 shows a sample letter that will be sent out to the different companies. An image showing the time frames assigned for each task that needs to be completed up until December of this year is shown in Figure.2, ending with the mid-term report. ✓

This type of project peaked my interest because of its relation to nano-structured materials and how they can be applied and used to improve life in general, but more specifically in the medical, dental and anatomical use. I enjoyed observing and benefitting from an x-ray diffraction machine in previous years of university and this is one of the machines that will be used to characterise the different surfaces on the burs. → good.

Two studies that stand out to me as having done similar research (C.F.M Borges, 1999) (W.Ahmed, 2009) involved coating the top of conventional diamond burs with an unbroken diamond film obtained through chemical vapour deposition and then comparing the surface of the bur with and without the film using a scanning electron microscope after conducting different erosion and cutting tests using different materials. This investigation was done to try to improve the longevity of dental burs without removing from their durability and to find a solution to the possible risk of small nickel ions breaking off while the bur is being used and entering the human body, this is the metal that is used to bind the shank to the diamond bur. Both studies concluded that a dental bur equipped with the diamond film through chemical vapour deposition had a strong probability of increasing bur lifespan while also preventing contamination from the binding metal. ✓

The literature review is too short.

The equipment that will be needed for this project include a scanning electron microscope to show an enhanced view of the burs surface allowing me to describe the surface at a level invisible to the naked eye. To determine the composition of the burs an x-ray diffraction machine will provide an assessment of the different elements within the dental bur. There will be at least 1 of each different type of bur made from the 3 different materials diamond, tungsten carbide and steel, the titanium and carbon nano-engineered films which are to be placed over the burs will also be manufactured to be used in this experiment. The cost of the materials required should be minimal as I will have access to the larger machines through the university and I'm not going to be buying the burs from dental labs but will be asking them if they are able to donate them to me. The cost of manufacturing will be small for the different nanostructures as only tiny amounts will be needed. Good.

→ A quotation should be added.

Burs are essentially used as drills therefore there is a minor risk of cutting one's self when handling them, this will mainly be a risk to me while I am conducting characterisation techniques on the burs. To reduce this risk the burs can be handled only via the shank or gloves will be worn while handling the tools. Risk assessments for the different machines include radiation from the x-ray diffraction machine and backscattered electrons and x-rays produced during the use of the scanning electron microscope. Detailed risk assessments can be found in the handbooks for both machines. ✓

Bibliography

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